Abhyas Physics - Grade 11

Chapter 1: Electrostatics and Electric Charges

1.1 Introduction to Electric Charges and Quantization

Matter is composed of atoms containing positively charged protons and negatively charged electrons. Charge is quantized in integral multiples of elementary charge e = 1.602 x 10^-19 C.

Coulomb’s Law states that force between two point charges is inversely proportional to r^2.

Chapter 1.2: Gauss’s Law and Electric Flux

Electric flux is the measure of the electric field passing through a given surface area.

Formula: Flux = Closed Integral of E . dA = Q_enc / Epsilon_0

Gauss’s law provides a powerful technique for calculating electric fields of symmetric charge distributions.

Applications include spherical shells, infinite plane sheets, and cylindrical conductors.

Chapter 1.3: Electric Potential & Capacitors

Electric potential is the work done per unit charge in bringing it from infinity to a point. Capacitance C = Q / V measures the ability of a conductor to store electric potential energy. Parallel plate capacitor capacitance is given by C = (Epsilon_0 * A) / d.